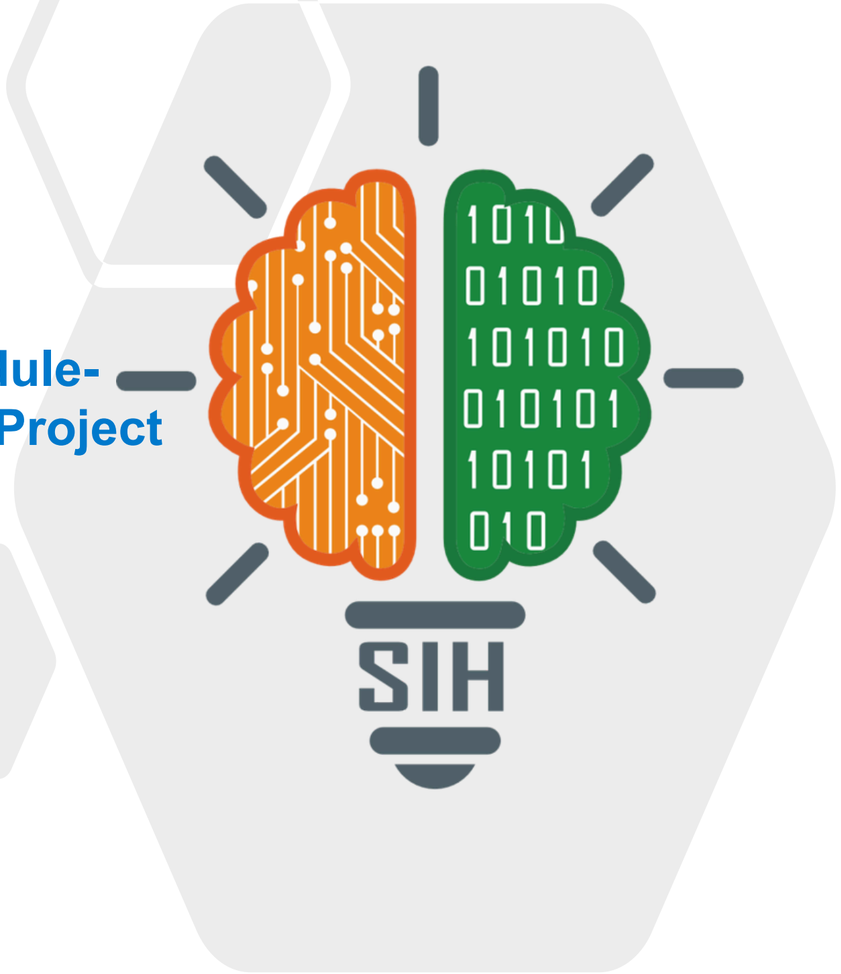
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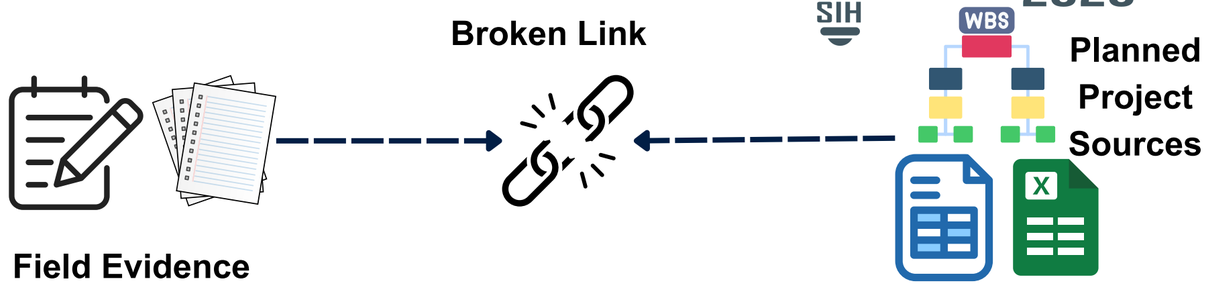
- **Problem Statement ID-** SIH26122
- **Problem Statement Title-** Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management: Real-Time Actual Progress Tracking (Planning-to-Execution Bridge)
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name-** AERES

AERES
EXECUTION INTELLIGENCE

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THE PROBLEM

- Infrastructure / civil Project schedules define detailed WBS execution (L5/L6) activities.
- But site progress comes through DPRs, spreadsheets, diaries and supervisors in different formats and terminology.
- Connecting “what happened” to “what was planned” becomes manual, slow and error-prone.



PROPOSED SOLUTION: AERES

Autonomous Infrastructure execution intelligence layer that connects field reality with the project schedule.

01 — CAPTURE & STRUCTURE

- Ingest DPRs, spreadsheets, site diaries & supervisor updates.
- Extract work, discipline, location & actual start/end.
- Convert different formats into structured execution events.

02 — RESOLVE & LINK

- Match field events to the correct L5/L6 schedule activity.
- Handle different terminology and work granularity.
- Use fuzzy, semantic & contextual matching.
- Flag ambiguous / unmatched / new activities for review.

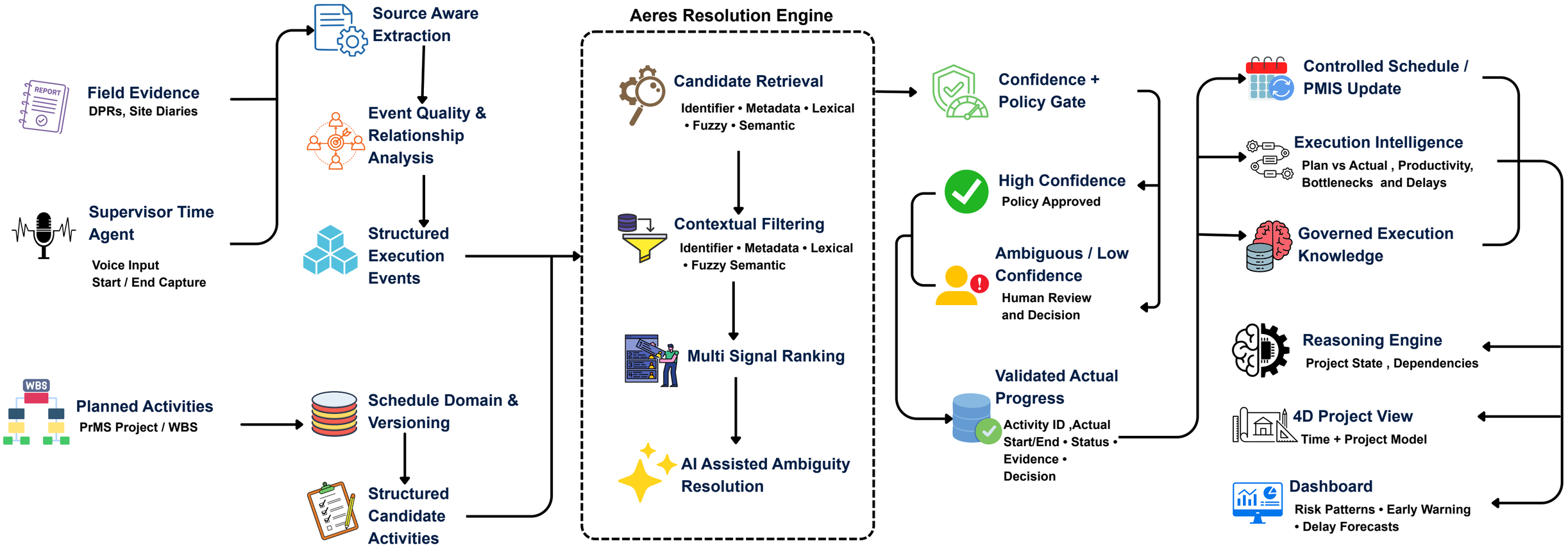
03 — GOVERN TRUSTED UPDATES

- Applies confidence- and evidence-based controls.
- Create validated actual-progress records.
- Keeps human oversight over critical updates.
- Generate controlled schedule / PMIS updates.

UNIQUE & ADDED FEATURES	REASON	VISUALIZE	FORECAST
	What should happen next? AI-driven next-action recommendations.	What is happening where? Dynamic 4D planned-vs-actual project view.	What could happen next? Execution-pattern-based risk & delay prediction.

TECHNICAL APPROACH

AERES LOGICAL SYSTEM ARCHITECTURE



TECH STACK	BACKEND	FRONTEND	DOCUMENT PROCESSING	AI / MATCHING	DATA	REASONING	OBSERVABILITY	DEPLOYMENT
	FastAPI Pydantic	Next.js	Docling PaddleOCR pandas/openpyxl	Sentence Transformers Structured LLM	PostgreSQL + pgvector	Rules Geometry Engine GLM-5.2	Langfuse OpenTelemetry	docker

AERES
EXECUTION INTELLIGENCE

FEASIBILITY AND VIABILITY

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TECHNICAL FEASIBILITY

- Hybrid Deterministic + AI Pipeline
- Multi source data Ingestion
- Evidence-Grounded, Human-Governed Decisions

OPERATIONAL FEASIBILITY

- Works with existing DPRs, spreadsheets and site reporting
- Planner Remains in Control
- Low-Friction, Incremental Adoption

IMPLEMENTATION FEASIBILITY

- Modular POC → MVP → Scale
- Incremental Capability Validation
- No Immediate PMIS Replacement

Evidence-Backed ✦ Scalable Data Moat ✦ Enterprise-Ready

VIABILITY

- **CLEAR ENTRY POINT**: Solves a specific, recurring planning–execution gap. Delivers value before advanced analytics are added
- **LOW ADOPTION BARRIER**: Uses reporting formats teams already produce. Adds intelligence without forcing a new project-control system
- **COMPOUNDING VALUE**: Every validated project record strengthens execution history. Enables progressively richer productivity, delay and risk intelligence

CHALLENGES

- Incorrect but Plausible Activity Matches
- Complex One-to-Many Work Relationships
- Trust & Adoption of AI Recommendations
- Inconsistent Evidence Across Sources

STRATEGIES

- Explicit Uncertainty & No Forced Matches
- Relationship-Aware Resolution & Exception Handling
- Human Validation Before Trusted Outcomes
- Evidence-Grounded Decision Traceability

IMPACT AND BENEFITS

Enhanced Schedule Integrity

Converts fragmented updates into a more **coherent view** of **actual execution**.

- Makes progress, delays and bottlenecks easier to analyse
- Preserves the context behind reported site activity

Non-Disruptive Interoperability

Adds an **intelligence layer** around existing project controls.

- Preserves established PMIS and planning workflows
- Enables gradual adoption without major process disruption

Accelerated Reconciliation

Automatically narrows ***field evidence*** to the ***most relevant schedule context***.

- Reduces repetitive manual activity mapping
- Speeds up progress reconciliation cycles

Frictionless Field Capture

Works with the **evidence** formats project teams already produce.

- Supports reports, spreadsheets, diaries and supervisor inputs
- Avoids forcing teams into entirely new reporting workflows

Consistent Project Intelligence

Validated outcomes **retain value** beyond a single reporting cycle.

- Builds reusable execution knowledge over time
- Creates a stronger foundation for future risk signals and forecasting

RESEARCH AND REFERENCES

References

- NIST AI Risk Management Framework: Generative AI Profile (NIST.AI.600-1) regarding confabulation and automation bias.
- Academic Journals: Zhan et al. (2025) on AI-specific risks in CPM; Singh & Hsieh (2026) on fragmented schedule data.
- Market Research: Grand View Research & Fortune Business Insights (2025-2034) sizing AI in Construction and AEC markets.
- Open Source Repositories: Docling, LangGraph, pgvector, and Langfuse for deterministic and stateful pipelines.

Research

- Data Fragmentation: Market reviews (e.g., Singh & Hsieh, 2026) repeatedly identify unstructured, disconnected field data as a major barrier, validating AERES's upstream ingestion strategy.
- Hallucination Risks: NIST and AEC research emphasize that confidently wrong outputs are critical risks, demanding our human-in-the-loop and confidence-scoring architecture.
- Competitive Whitespace: Industry research (like reports from Grand View Research) reveals that while broad enterprise platforms handle system-of-record breadth, a critical whitespace exists at the specialized translation layer between messy field language and L5/L6 schedule updates.

YOU CAN ACCESS OUR COMPLETE PRODUCT DETAILS AND RESEARCH HERE: AERES